Name: Gaurika Mittal

Batch: 15

Sap ID: 590034892

EXPERIMENT-7

Question 1

Problem Statement

Write a C program to find the **Curved Surface Area (CSA), Total Surface Area (TSA), and Volume of a Cone** when its radius and height are given.

Step 1: Problem Statement

Input

- Radius (r)
- Height (h)

Output

- Curved Surface Area
- Total Surface Area
- Volume

Formula

Slant Height = $\sqrt{(r \times r + h \times h)}$

Curved Surface Area = $\pi \times r \times$ Slant Height

Total Surface Area = $\pi \times r \times$ (r + Slant Height)

Volume = $(\pi \times r \times r \times h) / 3$

Take $\pi = 3.14$

Step 2: Test Case Table

Test Case	Inputs	Expected Outputs	Actual Outputs	Results
1	r=3 h=4	CSA=47.10 TSA=75.36 Volume=37.68	Same	Pass
2	r=5 h=12	CSA=204.10 TSA=282.60 Volume=314.00	Same	Pass
3	r=7 h=24	CSA=543.34 TSA=697.20 Volume=1230.88	Same	Pass

Step 3: Algorithm

1. Start
2. Declare variables r, h, l, csa, tsa, volume.
3. Read radius and height.
4. Calculate slant height using $\sqrt{(r \times r + h \times h)}$.
5. Calculate curved surface area.
6. Calculate total surface area.
7. Calculate volume.
8. Display CSA, TSA and Volume.
9. Stop.

Step 4: Flowchart

1. Start
2. Input Radius and Height
3. Calculate Slant Height
4. Calculate Curved Surface Area
5. Calculate Total Surface Area
6. Calculate Volume
7. Display CSA, TSA, Volume
8. Stop

Step 5: C Program

```
#include <stdio.h>
int main()
{
   float r, h, l, csa, tsa, volume;
```

```
    printf("Enter radius: ");
    scanf("%f", &r);
    printf("Enter height: ");
    scanf("%f", &h);
    l = sqrt(r * r + h * h);
    csa = 3.14 * r * l;
    tsa = 3.14 * r * (r + l);
    volume = (3.14 * r * r * h) / 3;
    printf("Curved Surface Area = %.2f\n", csa);
    printf("Total Surface Area = %.2f\n", tsa);
    printf("Volume = %.2f\n", volume);
    return 0;
}
```

Step 6: Compilation & Execution Evaluation

Test Case	Results
1	Pass
2	Pass
3	Pass

Step 7: Screenshots

```
Curved Surface Area = 549.50
Total Surface Area = 703.36
Volume = 1230.88
```

Question 2

Problem Statement

Write a C program to find the **Radius, Diameter and Area of a Circle** when its center and one point on the circle are given.

Step 1: Problem Statement

Input

Center coordinates:(x_1, y_1)

Point on circle:(x_2, y_2)

Output

- Radius
- Diameter
- Area

Formula

Radius = $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

Diameter = 2 × Radius

Area = π × Radius × Radius

Take π = 3.14

Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Results
1	(0,0) (3,4)	Radius=5 Diameter=10 Area=78.50	Same	Pass
2	(1,2) (4,6)	Radius=5 Diameter=10 Area=78.50	Same	Pass

3	(2,3) (5,7)	Radius=5 Diameter=10 Area=78.50	Same	Pass

Step 3: Algorithm

1. Start.
2. Declare x1, y1, x2, y2, radius, diameter, area.
3. Read center coordinates.
4. Read point coordinates.
5. Calculate radius.
6. Calculate diameter.
7. Calculate area.
8. Display radius, diameter and area.
9. Stop.

Step 4: Flowchart

1. Start
2. Input x_1, y_1
3. Input x_2, y_2
4. Calculate Radius
5. Calculate Diameter
6. Calculate Area
7. Display Radius, Diameter, Area
8. Stop

Step 5: C Program

```
#include <stdio.h>
#include <math.h>
int main()
{
    float x1, y1, x2, y2;
    float radius, diameter, area;
    printf("Enter center coordinates (x1 y1): ");
    scanf("%f %f", &x1, &y1);
    printf("Enter point coordinates (x2 y2): ");
    scanf("%f %f", &x2, &y2);
    radius = sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
    diameter = 2 * radius;
    area = 3.14 * radius * radius;
    printf("Radius = %.2f\n", radius);
    printf("Diameter = %.2f\n", diameter);
```

```
    printf("Area = %.2f\n", area);
    return 0;
}
```

Step 6: Compilation & Execution Evaluation